

**Authored Grade-9 Force and Laws of Motion
spine covering balanced/unbalanced for:
Orientation map**

Zero net force allows steady motion

Hard transition

Zero net force does not mean zero motion; it means velocity does not change, so rest and steady motion are both compatible with the same condition.

Governing relation

$$F_{\text{net}} = m a$$

The vector net external force on a chosen body equals its mass times its acceleration.

- $m > 0$.
- F_{net} and a refer to the same chosen body.
- All signed components use one declared axis.

Compact anchor

A 3 kg cart glides at 2 m/s on a frictionless track with no horizontal force on it. What is its velocity 5 s later?

$F_{\text{net}} = 0$, so $a = 0/3 = 0$ and the velocity is still 2 m/s: zero net force keeps the motion unchanged, it does not stop it.

Before you go on

One cart is at rest and another moves steadily in a straight line. For each, decide whether zero net force is compatible with the motion and explain the common rule.

Nonzero forces can cancel

Hard transition

Several nonzero forces can act on one body while their directed sum is zero; the motion responds to the net result rather than to the mere presence of forces.

Governing relation

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The vector net external force on a chosen body equals its mass times its acceleration.

- $m > 0$.
- F_{net} and a refer to the same chosen body.
- All signed components use one declared axis.

Compact anchor

A 2 kg box is pushed with 5 N to the left and 5 N to the right at the same time. What is its acceleration?

$F_{\text{net}} = 5 - 5 = 0$ N, so $a = 0/2 = 0$: two real, non-zero forces act, and the velocity does not change.

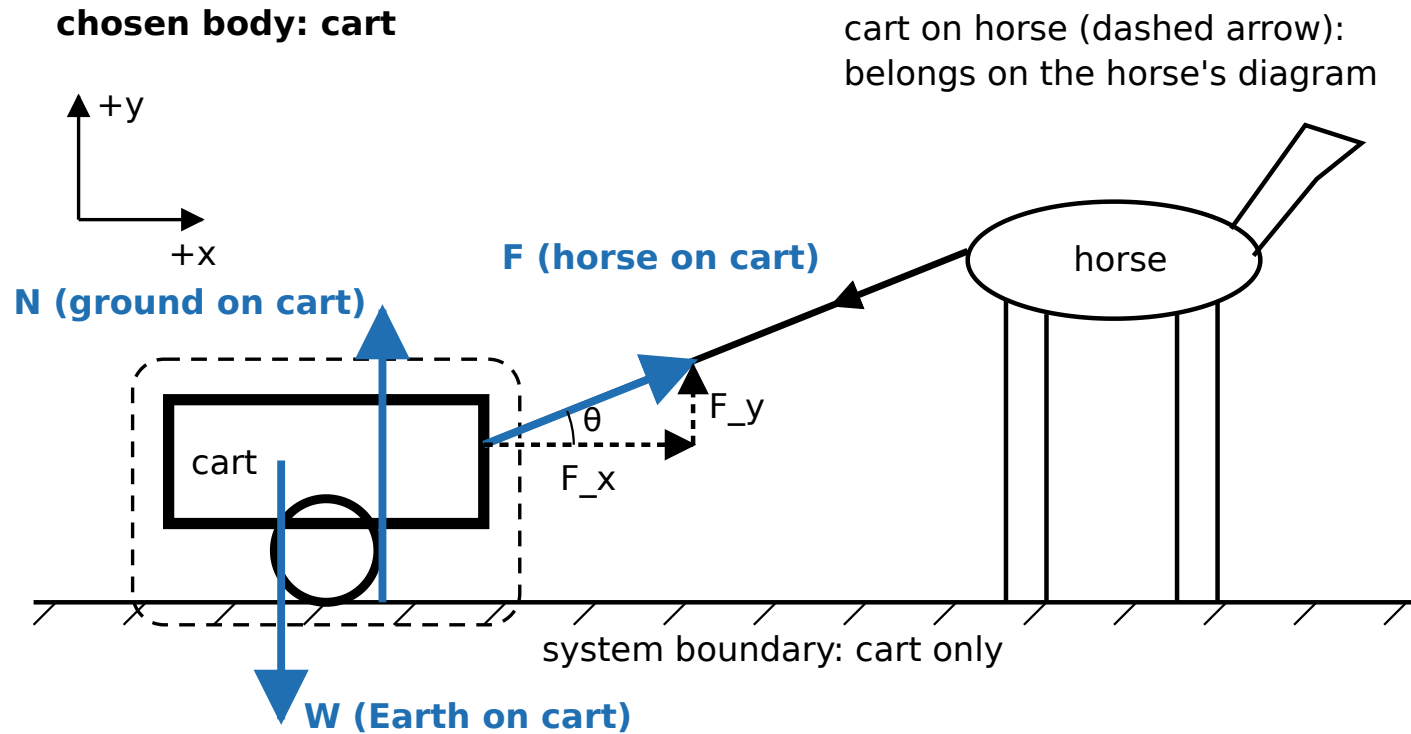
Before you go on

An object has two equal opposite forces acting on it. Explain what is zero and what is not, then state what motion change the net result predicts.

A free-body diagram belongs to one chosen body

Hard transition

A free-body diagram contains forces acting on one chosen body only; the equal-and-opposite partner of an interaction acts on the other body and belongs on that other body's diagram.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

Governing relation

$$F_{\text{net}} = m a$$

The vector net external force on a chosen body equals its mass times its acceleration.

- $m > 0$.
- F_{net} and a refer to the same chosen body.
- All signed components use one declared axis.

Compact anchor

A horse pulls a 200 kg cart with 600 N and road friction on the cart is 200 N. What is the cart's acceleration, and where does the cart's pull on the horse belong?

On the cart's diagram only: $F_{\text{net}} = 600 - 200 = 400 \text{ N}$, so $a = 400/200 = 2 \text{ m/s}^2$. The cart-on-horse partner belongs on the horse's diagram, not in this sum.

Before you go on

A person pushes a box. Describe the two contact-force partner arrows and state which one belongs on the box's free-body diagram.

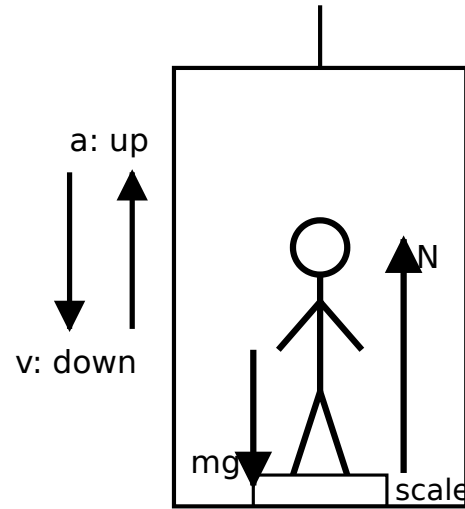
An accelerating observer needs a declared pseudo-force convention

Hard transition

A force-based account depends on the observer: a non-accelerating observer uses the physical interaction forces, while an accelerating observer must add the corresponding pseudo-force convention to describe the same observed motion consistently.

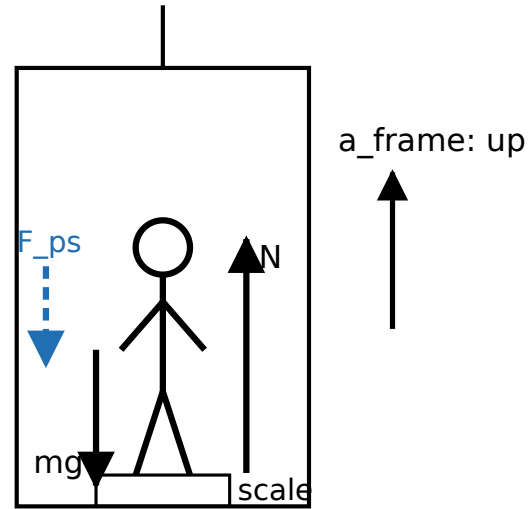
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 5

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

Governing relation

$$N - m g = m a_y$$

For a person that remains in contact with a horizontal scale in an inertial ground frame, with upward positive and g the positive magnitude of gravitational acceleration, the signed vertical force balance is $N - m g = m a_y$.

- The ground frame is treated as inertial and the positive vertical direction is upward.
- The person remains in contact with the scale, so $N \geq 0$.
- The person and scale share the stated vertical acceleration while contact persists.
- g is the positive magnitude of the downward gravitational acceleration.

$$F_{\text{pseudo}} = -m a_{\text{frame}}$$

In a one-axis frame accelerating with signed acceleration a_{frame} relative to an inertial frame, the pseudo-force assigned to a mass m is opposite the frame acceleration and has magnitude $m|a_{\text{frame}}|$.

- The selected observer frame has nonzero translational acceleration relative to the inertial frame.

- F_{pseudo} and a_{frame} use the same declared axis and sign convention.
- The mass is positive and treated as constant.
- The pseudo-force is a frame-dependent modelling term, not a physical interaction and not a Newton-third-law partner.

$$\tan(\theta) = |a_{\text{frame}}| / g$$

For a bob stationary relative to a horizontally accelerating vehicle, with a massless taut string and no other horizontal interaction, the equilibrium string tilt magnitude θ from vertical satisfies $\tan(\theta) = |a_{\text{frame}}|/g$ in the vehicle frame.

- The bob is stationary relative to the vehicle after transients, so its relative acceleration is zero.
- The vehicle has constant horizontal acceleration over the represented state.
- The string is massless and taut, and air resistance is neglected.
- θ is measured from the vertical; the bob tilts opposite the vehicle acceleration.

Compact anchor

A 2 kg bag sits on the floor of a van that accelerates forward at 3 m/s^2 . What pseudo-force does an observer riding in the van assign to the bag?

$F_{\text{pseudo}} = -m a_{\text{frame}} = -2 \times 3 = -6 \text{ N}$: 6 N backward, used only in the van's description and never paired with a physical force.

Before you go on

A bus accelerates forward and a loose object appears to move backward relative to the bus. Compare the roadside and bus-observer force descriptions without calculating any magnitude.

Net force, not any one force, sets acceleration

Hard transition

For a chosen body of positive mass in the Grade 9 inertial-frame model, acceleration points with the signed net external force and satisfies $F_{\text{net}} = m a$.

Governing relation

$$F_{\text{net}} = m a$$

The vector net external force on a chosen body equals its mass times its acceleration.

- $m > 0$.
- F_{net} and a refer to the same chosen body.
- All signed components use one declared axis.

Compact anchor

A 5 kg trolley has 20 N forward and 5 N backward acting on it. What is its acceleration?

$F_{\text{net}} = 20 - 5 = 15$ N forward, so $a = 15/5 = 3$ m/s² forward, in the direction of the net force.

Before you go on

A 4 kg body has a net external force of +12 N along a declared axis. Find its acceleration.

Friction opposes relative sliding at a contact

Hard transition

Friction is a contact force along the surface that opposes relative sliding, or the tendency for relative sliding, between the contacting surfaces; it is not defined by the object's overall velocity alone.

Governing relation

$$|f_s| \leq \mu_s N$$

For a contact that remains static, the static-friction magnitude adjusts to the value required by the tangential dynamics up to the limiting value $\mu_s N$; equality applies only at impending slip.

- The surfaces are in contact and have no relative sliding during the state being tested.
- N is obtained from the actual free-body dynamics, not assumed to equal mg .
- Static friction opposes the impending relative slip direction.
- Equality $|f_s| = \mu_s N$ is used only at impending slip.

Compact anchor

A 2 kg box on a rough floor ($\mu_s = 0.5$, $g = 10 \text{ m/s}^2$) is pushed sideways with 6 N and does not move. What friction acts on it?

6 N, opposite the push: rest needs friction to balance the push, and 6 N is within the limit $\mu_s N = 0.5 \times 20 = 10 \text{ N}$.

Before you go on

A box is placed on a conveyor belt moving to the right faster than the box. Before the box matches the belt speed, what direction is the friction force on the box?

Third-law partner forces act on different bodies

Hard transition

One interaction produces two partner forces of equal magnitude and opposite direction, but they act on different bodies and therefore do not cancel in a single body's net-force sum.

Governing relation

$$F_{\text{net}} = m a$$

The vector net external force on a chosen body equals its mass times its acceleration.

- $m > 0$.
- F_{net} and a refer to the same chosen body.
- All signed components use one declared axis.

Compact anchor

On smooth ice a 60 kg skater pushes a 30 kg skater with 90 N. What are their accelerations?

Each gets 90 N, on different bodies: the pushed skater has $a = 90/30 = 3 \text{ m/s}^2$ forward, the pusher $a = 90/60 = 1.5 \text{ m/s}^2$ backward. The pair never cancels, because each force is in a different body's F_{net} .

Before you go on

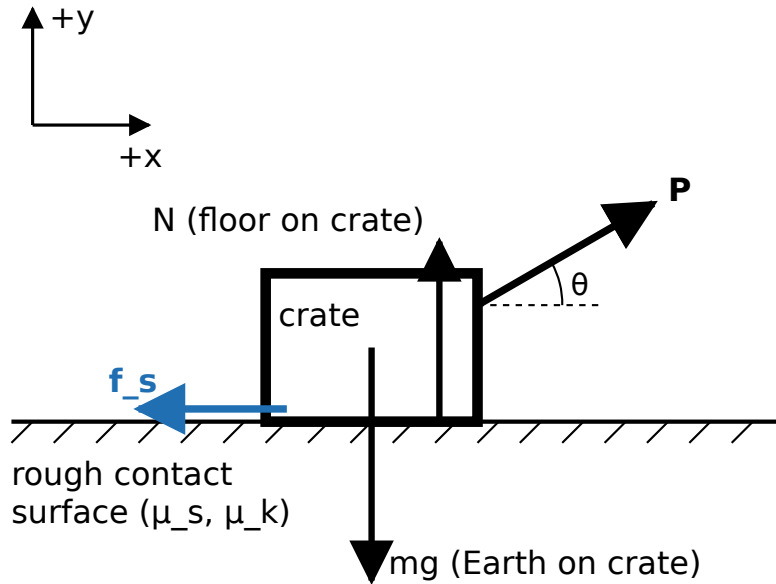
A swimmer pushes water backward. Identify the third-law partner force and state which body each force acts on.

Quantitative static and kinetic friction conditions

Hard transition

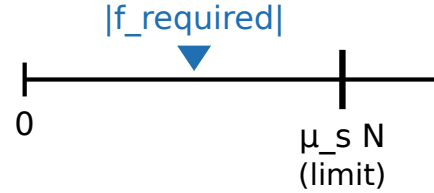
Static friction is a responsive contact force: solve for the friction required to prevent relative slip and compare it with the limiting value $\mu_s N$. Equality is a threshold condition, while the kinetic model is used only after sliding is established.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



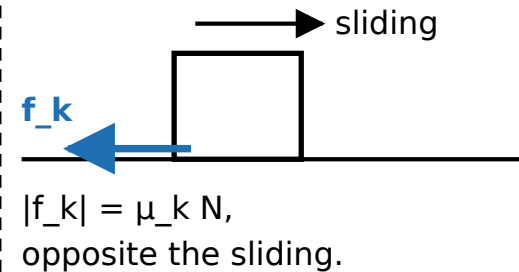
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

Governing relation

$$|f_s| \leq \mu_s N$$

For a contact that remains static, the static-friction magnitude adjusts to the value required by the tangential dynamics up to the limiting value $\mu_s N$; equality applies only at impending slip.

- The surfaces are in contact and have no relative sliding during the state being tested.
- N is obtained from the actual free-body dynamics, not assumed to equal mg .
- Static friction opposes the impending relative slip direction.
- Equality $|f_s| = \mu_s N$ is used only at impending slip.

$$|f_k| = \mu_k N$$

Once relative sliding at the contact is established, the bounded dry-sliding model uses kinetic-friction magnitude $\mu_k N$, directed opposite the relative sliding.

- Relative sliding between the two surfaces is established.
- N is obtained from the actual free-body dynamics.
- The friction direction opposes the relative sliding at the contact.
- The simple dry-sliding coefficient model is being used.

Compact anchor

A 2 kg block on a floor with $\mu_s = 0.5$ and $\mu_k = 0.3$ is pulled horizontally with 14 N ($g = 10 \text{ m/s}^2$). Does it stay put, and what is its acceleration?

$N = 20 \text{ N}$, so static friction can give at most 10 N; $14 \text{ N} > 10 \text{ N}$, so it slides. Then $f_k = 0.3 \times 20 = 6 \text{ N}$ and $a = (14 - 6)/2 = 4 \text{ m/s}^2$.

Before you go on

A contact has known normal reaction N and coefficient μ_s . The no-slip force equation requires tangential friction f_{required} . State the test for whether static contact is possible, when equality with $\mu_s N$ is justified, and what changes once sliding is established.

Connected bodies and a justified common acceleration

Hard transition

Bodies share an acceleration component only when a physical constraint keeps their relative separation fixed along that coordinate; the shared acceleration can link separate Newton-II equations, while internal partner forces cancel only if the bodies are deliberately treated as one system.

Governing relation

$$a_A = a_B$$

When an active connection keeps the relative separation of bodies A and B fixed along one shared coordinate and both use the same positive sense, their acceleration components along that coordinate are equal.

- The connection/contact actively keeps the relevant relative separation fixed.
- Both acceleration components use the same coordinate direction and sign convention.
- The constraint has not gone slack, separated, or otherwise ceased to apply.

Compact anchor

A 30 N push moves a 4 kg block and a 2 kg block together across a smooth floor, the first pressing on the second. What is their shared acceleration?

The maintained contact fixes their separation, so $a_A = a_B = 30/(4 + 2) = 5 \text{ m/s}^2$.

Before you go on

Two bodies are constrained to keep a fixed separation along one straight coordinate. State the acceleration relation, write the form of the two body equations, and explain when their interaction force can disappear from an equation.

Ideal-string tension belongs to each body-specific FBD

Hard transition

For one taut massless ideal string, with any redirection treated as frictionless and massless, the tension magnitude can be represented by one T throughout that same string; each body still receives its own tension force along the local string direction.

Governing relation

$$T_A = T_B = T$$

For one taut massless ideal string, with any direction-changing pulley/support treated as frictionless and massless in this bounded model, the tension magnitude is uniform throughout that same string.

- The same taut string connects the stated segments.
- The string is massless and inextensible in the ideal model.
- Any pulley/redirection used here is frictionless and massless and only changes string direction.
- No nonideal feature is introduced that would require different tension magnitudes.

Compact anchor

A light string over a frictionless, light pulley joins two bodies. The string pulls one of them with 8 N. How hard does it pull the other?

Also 8 N: under the ideal model $T_A = T_B = T$, and each pull acts on its own body along its own string segment.

Before you go on

Two bodies are attached to the same taut ideal string, redirected only by a frictionless massless fixed pulley. State the tension relation and explain where the two tension forces appear.

One fixed-pulley string gives one length constraint

Hard transition

For one taut inextensible string over one fixed ideal pulley, the two changing straight segment lengths add to a constant; with coordinates measured away from the pulley this gives $y_A + y_B = \text{constant}$ and therefore $a_A + a_B = 0$.

Governing relation

$$y_A + y_B = L_{\text{free}} = \text{constant}; \quad a_A + a_B = 0$$

For one taut inextensible string over one fixed ideal pulley, with both coordinates measured away from the pulley along the two straight segments, the variable segment lengths sum to a constant and the signed accelerations sum to zero.

- There is one fixed pulley and one taut inextensible string.
- Only the two declared straight segments change length; the wrapped/fixed portions are constant.
- Both coordinates are measured away from the pulley along their local string segments.
- The pulley itself does not translate.

Compact anchor

Over a fixed pulley, y_A and y_B are measured away from the pulley along the string. A accelerates away from the pulley at 3 m/s^2 . What is a_B ?

$y_A + y_B$ stays constant, so $a_A + a_B = 0$ and $a_B = -3 \text{ m/s}^2$: B accelerates toward the pulley at 3 m/s^2 .

Before you go on

One taut inextensible string passes over one fixed ideal pulley, with y_A and y_B measured away from the pulley along the two hanging segments. Write the length relation and its acceleration consequence.